

A Toeplitz-Algebra Counterexample to a C^* -Distance Formula Question

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Status. `counterexample_likely_valid`.

1 Source Question

The source paper is Grover–Singla, *Birkhoff-James orthogonality and applications: A survey*, arXiv:2005.07399 [1]. On pages 15–16, after recalling Rieffel’s distance formula, the authors ask whether the following formula holds. If $\mathcal{A}$ is a unital irreducible C^* -algebra, $a \in \mathcal{A}$, and $\mathcal{B}$ is a unital C^* -subalgebra of $\mathcal{A}$, is it true that

$$2 \operatorname{dist}(a, \mathcal{B}) = \sup\{\|au - ua\| : u \in \mathcal{A}, u \text{ unitary, and } bu = ub \text{ for all } b \in \mathcal{B}\} \quad (1)$$

The source states that it is open whether (1) is true.

for all $b \in \mathcal{B}$ and $\text{dist}(a, \mathcal{B}) = \frac{1}{2} \|\pi(a)U - U\pi(a)\|$.

By Theorem 4.2, we obtain

$$2 \text{ dist}(a, \mathcal{B}) = \max\{\|\pi(a)U - U\pi(a)\| : U \in \mathcal{B}(\mathcal{H}), U = U^*, U^2 = 1_{\mathcal{B}(\mathcal{H})}, \\ (\mathcal{H}, \pi, \xi) \text{ is a cyclic representation of } \mathcal{A}, \text{ and} \\ \pi(b)U = U\pi(b) \text{ for all } b \in \mathcal{B}\}.$$

Looking at the last expression and (4.2), it is tempting to conjecture that if $\mathcal{A}$ is a unital irreducible C^* -algebra, $a \in \mathcal{A}$ and $\mathcal{B}$ is a unital C^* -subalgebra

of $\mathcal{A}$, then

$$(4.3) \quad 2 \text{ dist}(a, \mathcal{B}) = \sup\{\|au - ua\| : u \in \mathcal{A}, u \text{ is a unitary element, and } bu = ub \\ \text{ for all } b \in \mathcal{B}\}.$$

We note that it is not possible to prove (4.3) by proceeding along the lines of the proof of Theorem 4.1 given in [14], which uses (4.1). In particular, the following does not hold true in $M_n(\mathbb{C})$:

$$\text{dist}(A, \mathcal{B}) = \max\{|\langle Ax|y \rangle| : \|x\| = \|y\| = 1 \text{ and } y \perp Bx \text{ for all } B \in \mathcal{B}\}.$$

For example, take $A = 1_{M_n(\mathbb{C})}$ and $\mathcal{B} = \{X \in M_n(\mathbb{C}) : \text{tr}(X) = 0\}$. Then $1_{M_n(\mathbb{C})}$ is orthogonal to $\mathcal{B}$. Now if the above is true, then we would get unit vectors x, y such that $|\langle x|y \rangle| = 1$ and $\langle Bx|y \rangle = 0$ for all $B \in \mathcal{B}$. Let $P = xy^*$. Then $\text{rank } P = 1$ and $\text{tr}(BP) = 0$ for all $B \in \mathcal{B}$. But $\text{tr}(BP) = 0$ for all $B \in \mathcal{B}$ gives $P = \lambda 1_{M_n(\mathbb{C})}$ for some $\lambda \in \mathbb{C}$ (see [32, Remark 3]), which contradicts the fact that $\text{rank } P = 1$. So this approach to prove (4.3) does not work. However it would be interesting to know if (4.3) is true or not. This is an open question.

The above example contradicts Theorem 5.3 of [101], which says that

2 Proof Intuition

The right hand side of (1) only sees unitaries in the relative commutant $\mathcal{B}' \cap \mathcal{A}$. If $\mathcal{B}$ is already irreducible inside $\mathcal{A}$, this relative commutant can be only the scalar multiples of the identity. Then the right hand side is forced to be zero for every a . A proper irreducible subalgebra therefore gives a counterexample as soon as one chooses $a \notin \mathcal{B}$.

The standard Toeplitz algebra provides a very concrete example. It is generated by the unilateral shift S on $\ell_2(\mathbb{N})$, has scalar commutant, and is a proper subalgebra of $B(\ell_2)$. A simple diagonal unitary $De_n = (-1)^n e_n$ lies outside it because Toeplitz-algebra elements have compact commutator with S , whereas $[D, S]$ is noncompact.

3 Counterexample

Theorem 1. *Formula (1) is false.*

Proof. Let $\mathcal{H} = \ell_2(\mathbb{N})$ with orthonormal basis $(e_n)_{n \geq 0}$, and let $\mathcal{A} = B(\mathcal{H})$. Then $\mathcal{A}$ is a unital irreducible C^* -algebra. Let $S \in B(\mathcal{H})$ be the unilateral shift, $Se_n = e_{n+1}$, and set

$$\mathcal{B} = C^*(S) \subset B(\mathcal{H}).$$

This is the unital Toeplitz algebra. Define a diagonal unitary D by

$$De_n = (-1)^n e_n \quad (n \geq 0).$$

We first compute the relative commutant. If $T \in B(\mathcal{H})$ commutes with every element of $\mathcal{B}$, then T commutes with both S and S^* . Since $S^*e_0 = 0$, we have

$$S^*Te_0 = TS^*e_0 = 0,$$

so $Te_0 \in \ker S^* = \mathbb{C}e_0$. Write $Te_0 = \lambda e_0$. Because T commutes with S ,

$$Te_n = TS^n e_0 = S^n Te_0 = \lambda e_n$$

for every n . Hence $T = \lambda I$. Therefore

$$\mathcal{B}' \cap \mathcal{A} = \mathbb{C}I.$$

The only unitary elements of $\mathcal{A}$ commuting with all of $\mathcal{B}$ are thus λI , $|\lambda| = 1$. For such a unitary,

$$D(\lambda I) - (\lambda I)D = 0.$$

Consequently the right hand side of (1), with $a = D$, is 0.

It remains to show that the left hand side is positive. We prove $D \notin \mathcal{B}$. Let

$$\mathcal{E} = \{X \in B(\mathcal{H}) : XS - SX \text{ is compact}\}.$$

This is a norm-closed unital algebra: if $X, Y \in \mathcal{E}$, then

$$(XY)S - S(XY) = X(YS - SY) + (XS - SX)Y,$$

which is compact, and norm-closedness follows because the compact operators are norm closed. Also $S \in \mathcal{E}$, and

$$S^*S - SS^* = I - SS^*$$

is the rank-one projection onto $\mathbb{C}e_0$, so $S^* \in \mathcal{E}$. Since $C^*(S)$ is the norm-closed unital algebra generated by S and S^* , it follows that

$$C^*(S) \subseteq \mathcal{E}.$$

But

$$(DS - SD)e_n = De_{n+1} - S((-1)^n e_n) = (-1)^{n+1} e_{n+1} - (-1)^n e_{n+1} = -2(-1)^n e_{n+1}.$$

Thus $DS - SD$ sends the orthonormal basis (e_n) to vectors of constant norm 2. It is not compact. Hence $D \notin \mathcal{E}$, and therefore $D \notin C^*(S) = \mathcal{B}$.

Since $\mathcal{B}$ is norm closed and $D \notin \mathcal{B}$, we have $\text{dist}(D, \mathcal{B}) > 0$. Thus the left hand side of (1) is $2 \text{dist}(D, \mathcal{B}) > 0$, while the right hand side is 0. This contradicts (1). $\square$

4 Verification Notes

The proof is analytic and does not rely on computation. The included script `code/toeplitz_sanity_check.py` verifies the finite-section shadow of the construction: the commutator of the alternating diagonal with the shift has stable norm 2, matching the noncompactness mechanism in the proof.

Novelty checks were bounded as follows. The run indexes were searched for 2005.07399, Birkhoff-James, distance formula, C*-subalgebra, and Toeplitz. Web searches on 2026-06-19 for exact and close variants of (1) found the source arXiv paper but no later paper explicitly resolving the question.

5 Limitations and Review Recommendation

This counterexample disproves formula (1) as stated when “unital irreducible C^* -algebra” is interpreted in the standard represented sense, allowing $\mathcal{A} = B(\ell_2)$. It does not address possible repaired formulas with additional hypotheses, for example requiring a sufficiently large relative commutant or replacing unitaries in $\mathcal{A} \cap \mathcal{B}'$ by unitaries in a cyclic representation as in Rieffel’s theorem.

Human review should focus on the intended meaning of “irreducible” in the source question and on the elementary Toeplitz-commutant/properness checks.

References

- [1] P. Grover and S. Singla, *Birkhoff-James orthogonality and applications: A survey*, arXiv:2005.07399, 2020.